Lennox Sama Keeble

Curriculum Vitae

Last updated: September 25, 2026.

Education

Wake Forest University **2026 - 2031**
Doctor of Philosophy (Ph.D.) in Physics. Supervisor: Professor Alejandro Cárdenas-Avendaño. *Winston-Salem, USA*

University of Cambridge **2025 - 2026**
Master of Advanced Study (MASt) in Theoretical Physics (Part III of the Mathematical Tripos). *Cambridge, UK*
Supervisor: Dr. Lorenzo Gavassino.

Princeton University **2020 - 2025**
Bachelor of Arts (B.A.) in Physics. Supervisor: Professor Frans Pretorius. *Princeton, USA*

Publication Summary

- **7** (submitted/accepted) refereed papers; **5** as (co-)lead author.
- **Papers by research area:** relativistic hydrodynamics (**2**); black hole imaging (**2**); extreme mass-ratio inspirals (**2**); black hole perturbation theory (**1**).
- Links to academic metrics: Google Scholar, INSPIRE.

Collaboration Memberships

- Laser Interferometer Space Antenna (LISA) Consortium Core Member (2026 –).
- Simons Collaboration on Black Holes and Strong Gravity (2026 –).

Honours, Scholarships & Awards

- Cambridge Trust Master's Studentship, University of Cambridge, 2025.
- Academic Futures Graduate Scholarship, University of Oxford, 2025. (Declined.)
- Allen G. Shenstone Prize in Physics, Princeton University, Department of Physics, 2025.
- Manfred Pyka Memorial Prize, Princeton University, Department of Physics, 2022.

Service

- Refereeing activities for Physical Review D (PRD).

Invited Talks, Conferences & Workshops

Bold **numbers** indicate a contributed talk; bold **text** indicates an invited talk.

8. IGWN-PI School on Gravitational Waves, Perimeter Institute for Theoretical Physics, 2026.
7. Relativistic Astrophysics Summer School, Caltech, 2025.
6. Nuclear Physics of MultiMessenger Mergers (NP3M) Summer School, Indiana University, 2025.
5. Los Alamos Astrophysics Distinguished Seminar Series, Center for Nonlinear Studies, Los Alamos National Laboratory, 2025.
4. Black Hole Mimickers: From Theory to Observation, Princeton Center for Theoretical Science & Princeton Gravity Initiative, 2025.
3. CU^2MIP Undergraduate Conference, College Park, USA, 2024.
2. American Physical Society April Meeting, 2024.
1. Taking it to the extreme: Symmetries and dynamics of extremal black holes, Princeton Center for Theoretical Science & Princeton Gravity Initiative, 2024.

Professional Experience

Investment Associate Intern — **Summer 2024**
Bridgewater Associates — *Westport, Connecticut, USA*

Summer Business Analyst — **Summer 2022**
McKinsey & Company — *San Francisco, California, USA*

Miscellaneous

Programming Languages: julia., Python, Mathematica & Slurm.
Web Links: Webpage, GitHub.

List of Publications

† indicates first-authored papers or alphabetical ordering.

Refereed (submitted/accepted)

[7] *Quadratic Gravitational-Wave Scattering by Kerr Black Holes.*
† **L. S. Keeble**, H. Zhu, L. E. Kidder, H. P. Pfeiffer, and M. A. Scheel.
arXiv: 2609.25522 (gr-qc).

[6] *A Systematic Study of Resonance-Driven Flux Modifications in Extreme-Mass-Ratio Inspirals.*
E. Levati, **L. S. Keeble**, and A. Cárdenas-Avendaño.
arXiv: 2609.25392 (gr-qc).

[5] *Radial oscillations of viscous stars.*
† **L. S. Keeble** and J. Redondo-Yuste.
Physical Review D 113, 124058 (2026) arXiv: 2603.23622 (gr-qc).

[4] *First-order viscous relativistic hydrodynamics on the two-sphere.*
† **L. S. Keeble** and F. Pretorius.
Physical Review D 112, 124034 (2025) arXiv: 2508.20998 (gr-qc).

[3] *Estimating high-order time derivatives of Kerr orbital functionals.*
† **L. S. Keeble** and A. Cárdenas-Avendaño.
Physical Review D 112, 084048 (2025) arXiv: 2508.08888 (gr-qc).

[2] *Inferring black hole spin from interferometric measurements of the first photon ring: A geometric approach.*
† **L. S. Keeble**, A. Cárdenas-Avendaño, and D. C. M. Palumbo.
Physical Review D 111, 103042 (2025) arXiv: 2502.20312 (astro-ph.HE).

[1] *Assessing the impact of instrument noise and astrophysical fluctuations on measurements of the first black hole photon ring.*
A. Cárdenas-Avendaño, **L. Keeble**, and A. Lupsasca.
Physical Review D 109, 124052 (2024) arXiv: 2404.01083 (gr-qc).

White Papers and Conference Proceedings

[1] *The Black Hole Explorer: photon ring science, detection, and shape measurement.*
A. Lupsasca, A. Cárdenas-Avendaño, D. C. M. Palumbo, M. D. Johnson, S. E. Gralla, D. P. Marrone, P. Galison, P. Tiede, and **L. Keeble**.
Proc. SPIE Int. Soc. Opt. Eng. 13092, 130926Q (2024) arXiv: 2406.09498 (gr-qc).